Matrices 1

Consolidation

7.
   a.
$$\begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & 1 & 2 & -1 \end{pmatrix}$$
   b.
$$\begin{pmatrix} 0 & 7 & 6 & 1 \\ 0 & -5 & -4 & -1 \end{pmatrix}$$

8.
   a.
$$A = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$
$$B = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$$
$$C = \begin{pmatrix} 6 \\ -9 \end{pmatrix}$$
$$D = \begin{pmatrix} 6 \\ -11 \end{pmatrix}$$
   b.

9. $\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

10.
   a.
$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
   b.
$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

11. a. 150° anticlockwise
    b. 143° anticlockwise

12. a. reflection in x-axis
    b. clockwise rotation of 45°
    c. shear factor of 2 in the x-axis.
    d. rotation of 90 about x-axis in 3 dimensions.

13.
   a.
$$AB = \begin{pmatrix} 4 & 1 \\ -9 & 7 \end{pmatrix}$$
   b.
$$BA = \begin{pmatrix} 3 & -1 & 3 \\ 1 & 8 & -9 \\ 8 & 4 & 0 \end{pmatrix}$$

Exam Questions

1. a.
$$\begin{aligned} AB &= \begin{pmatrix} (1 \cdot -1 + 2 \cdot 2) & (2 \cdot 1 + 2 \cdot -1) \\ (2 \cdot -1 + 1 \cdot 2) & (2 \cdot 2 + 1 \cdot -1) \end{pmatrix} \\ &= \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \\ &= 3 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

   b.
$$\begin{aligned} EF &= \begin{pmatrix} 2k + k^3 \\ 6 \end{pmatrix} \\ &= \begin{pmatrix} -2k \\ 6 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \therefore 2k + k^3 &= -2k \\ k^3 + 4k &= 0 \\ k(k^2 + 4) &= 0 \\ \therefore k &= 0 \text{ or } 2i \text{ or } -2i \end{aligned}$$

2. a. A shear in the x direction with shear factor 5.

   b.
$$Q = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

   c.
$$\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 0 & 1 \end{pmatrix}$$

3. $M = \begin{pmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$

   $\begin{pmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & \frac{4}{5} & -\frac{3}{5} & \frac{1}{5} \\ 0 & \frac{3}{5} & \frac{4}{5} & \frac{7}{5} \end{pmatrix}$